

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 8
 - Compactness

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Warm-up Challenge

Problem

Let $\vec{a} \in \mathbb{R}^n$.

Prove that the singleton set $\{\vec{a}\}$ is closed.

Point set topology

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- kinds of sets: open, closed, connected, component, compact
- kinds of functions: continuous, homeomorphism

Answers must be in terms of open sets

Open covers

Definition

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- $\{(0, 1/2), (1/3, 2)\}$ is an open cover of the interval $(0, 1]$
- $\{(\frac{1}{n}, \frac{n+1}{n}) : n \in \mathbb{Z}_+\}$ is an open cover of $(0, 1]$

Lindelöf Covering Theorem

Theorem (Lindelöf Covering Theorem)

Let $A \subseteq \mathbb{R}^n$ be a set and suppose $\{U_i : i \in I\}$ is an open covering of A . Then there exists a countable subcover $\{U_j : j \in J\}$.

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Can we do better than this?

Definition

A set $A \subseteq \mathbb{R}^n$ is called **compact** if every open cover of A has a *finite* subcover.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

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Suppose that $\{U_i : i \in I\}$ is an open cover of X .

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This means that $\vec{x} \in \bigcup_{i \in I} U_i$.

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Therefore $\vec{x} \in U_j$ for some $j \in I$.

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It follows that $\{U_j\}$ is a subcover of A consisting of a single set.

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Thus every open cover has a finite subcover, making A compact!

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Suppose A is a compact set.

Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = B(\vec{0}; i)$.



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Therefore $\vec{x} \in B(\vec{0}; N)$ and it follows $\vec{x} \in \bigcup_{i \in I} U_i$.

Thus $\mathbb{R}^n = \bigcup_{i \in I} U_i$, and since $A \subseteq \mathbb{R}^n$, we see $A \subseteq \bigcup_{i \in I} U_i$.

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Explain what it means that $\{U_i : i \in I\}$ has a finite subcover.

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Solution

There is a finite subset $\{i_1, i_2, \dots, i_m\} \subseteq I$ with $A \subseteq \bigcup_{k=1}^m U_{i_k}$.

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This shows A is bounded.

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Suppose that A is compact and let $\vec{x} \in \mathbb{R}^n \setminus A$.

We will show $\vec{x}$ cannot be an adherent point of A and therefore A contains all of its adherent points.

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Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = \mathbb{R}^n \setminus \overline{B}(\vec{x}; 1/i)$. □

Challenge

Problem

Prove that $\bigcup_{i \in I} U_i = \mathbb{R}^n \setminus \{\vec{x}\}$.

Challenge

Problem

Explain why $\{U_i : i \in I\}$ is an open cover of A .

Challenge

Problem

Show that if $\{U_i : i \in I\}$ has a finite subcover, then $\vec{x}$ can't be an adherent point of $\vec{A}$.